1. Surface Properties

(a) Air Permeability

* Defn + Short note

Q/A Air porosity :- Ratio of airspace to the total volm of the fabric expressed as percentage.

— Sinkle

This is an calculated value based on an estimation of the volm of the component fibres and the estimation of the volm of the fabric from measurement of length, width and thickness.

Let, S = Total volm of the sample in cc

F = Total volm of the fibre in cc

∴ Air porosity, $P = \frac{S - F}{S} \times 100$

Sinkle point out that, the type of finish given to a fabric can have a considerable effect on permeability even through porosity may remain same.

Q/1 Air resistance :- It is the time in seconds for 1 cm^3 of air pass through 1 cm^2 of the fabric under a pressure head of 1 cm of water.

Air resistance is the reciprocal of air permeability. If more time need to pass 1 cm^3 air means more air resistance and less air permeability.

Q/1 Air permeability :- The rate of air flow passing ~~through~~ perpendicularly through a known area under a prescribed air pressure differential beth two surface of a material.

Mathimatically, it is the voim of air measured in cc passed/second through 1 cm^2 of fabric under a pressure head of 1 cm of water. Units of air permeability are - mm/s, $ft^3/ft^2/min$, $cm^3/cm^2/s$ etc.

*** Air permeability $\propto \frac{1}{\text{Air resistance}}$

* Three principle to measure air permeability :-

- Defn of air permeability

I. Measurement of time required to pass the given volm of air through a given area of fabric.

↓

* Used to measure air resistance.

II. Measurement of pressure differential required for air to pass at a given constant volm rate through given fabric area.

III. Measurement of volm rate of flow of air through a given fabric area under a given pressure differential.

↓

* Used to measure air flow.

↓

* Most of the instrument are working on this principle.

- ASTM D737-96

(b) Abration resistance

* Defn + short note

Abration :- Physical destruction of fabric, yarns and fibres resulting from the rubbing of textile surface over another surface.

- Example - Friction between body and garments

Abrasion occus during washing the garments and during wearing.

* Types of abration :-

- Defn of abration. There are 3 types of abrasion, they are -

(1) Plane or flat abrasion :- A flat area of material is abraded through surface rubbing.

- Example - Abrasion beth bed sheet and body fabrics.

(2) Flex abrasion :- Rubbing is accompanied by flexing and bending or folding.

- Example - Abrasion due to knee on the garments.

(3) Edge abrasion :- Occurs at pleats, collar, folds, cuffs, cardings, hemp, pocket, bottom edge of parts etc. This type of abrasion occurs diagonally.

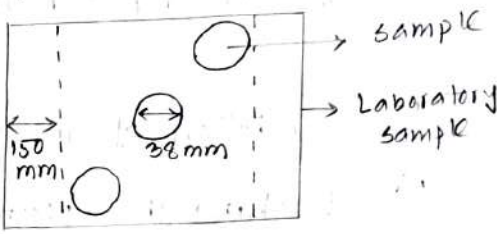
* Factors consider for abrasion tests :-

(1) Condition of the specimen (Standard atmosphere)
(2) Backing the specimen (Hardness)
(3) Tension on the specimen (Variation on tension neglected)
(4) Cleanliness of the specimen and instrument (substances act as lubricant)
(5) Choice of testing instrument (Acc to testing desire, types).
(6) Choice of abrasion motion (Reciprocating, rotary, multidimentional)
(7) ~~Choice of~~ Direction of abrasion (Angles to warp and weft direction
(8) Choice of abradant (vary abrasion)
(9) The pressure betn abradant and specimen (affected)
(10) The end point of the test. (As soon as hole appers)

Sample preparation → Abrasion resistance test → Report

* Sample preparation :-

(1) At least 3 samples are cut from laboratory sample diagolally so that, every sample's warp and weft are in different direction.



(2) Dimension of specimen/sample

- Specimen = 38 mm dia
- Abradant = 140 mm dia
  Abradant felt = 140 mm (wool/nonwooven) dia
- Foam = 38 mm dia

(3) Conditioning :

- Temperature = ~~(60±2)°C~~ (25±2)°C
- RH % = (65±2)%
- Time = 24 hours usually

* Testing procedures :-

i. 4 specimens, each of 38 mm in diameter are cut by using appropiate cutter.

ii. Than they are mounted into the specimen holders with a circle of standrad foam backing behind the fabric. The specimen must be ~~mounting~~ placed flat against the mounting block.

iii. The test specimen holders with the specimens are placed on the abradant.

iv. A spindle is inserted through the top plate and the correct weight (12 kPa or 9kPa) is placed on top of this.

v. A standard abradant should be replaced at the start of each test and also replaced after each 50,000 cycles if the test is continued beyond this number.

vi. While the abradant is replaced it is held flat by a weight and the retaining ring is tightened.

vii. Behind the abradant there is a standard backing felt which is replaced at longer interval.

viii. After appropiate clamping of the specimen on the holders, required number of cycles of abrasion is applied.

ix. Finally after testing, need to calculate strength or weight looss and asses the apperence changes, by the equn,

$$\text{Weight or strength loss } \% = \frac{A-B}{A} \times 100\%$$

where,

A = Wt or strength, before abration

B = " after "

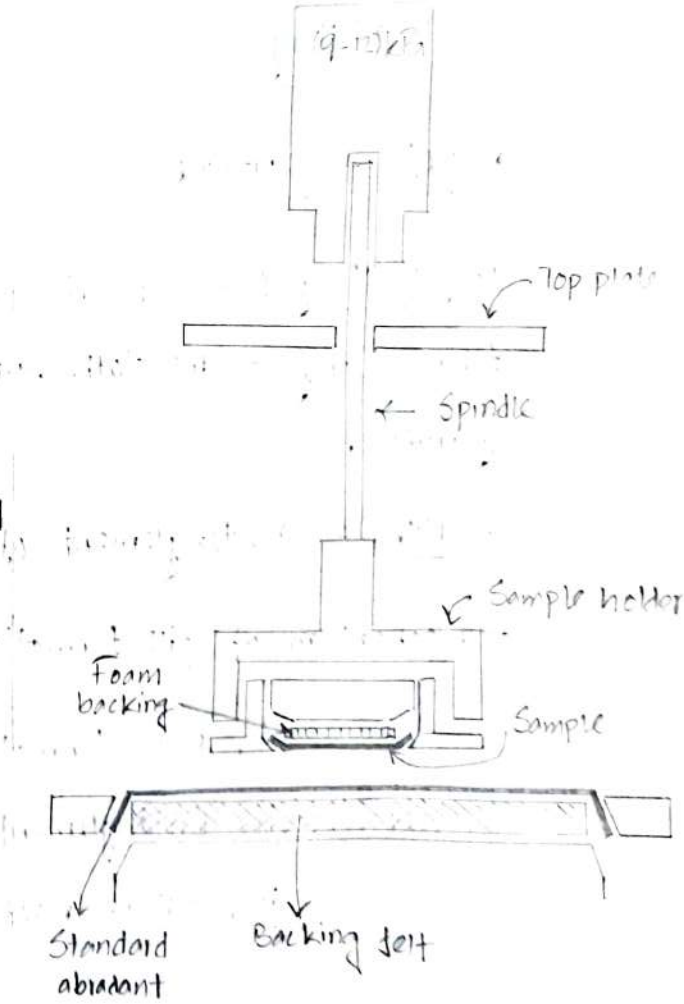


Fig:- One station of a Martindale abrasion tester

* Report :- i. The test is conducting according to ISO-12947 to determine abrasion resistance using Martindale tester.

ii. Types of material tested (Wooven).

iii. There are 4 no of reading in both warp and weft way

iv. No test is rejected.

© Pilling

* Defn + shortnote

Pilling :- Fabric surface fault in which pills of entangled fibre cling to the cloth surface, giving a bad apparence to the garment.

* Pills - Entanglement of loose fibre appear on fabric surface.

- Mechanism of pill formation have 3 distinct stages -

i. Fuzz formation (Due to mechanical action)

ii. Entanglement

iii. Pill wear off (Under continued mechanical action)

* Way to reduce pilling :-

→ Physical processes
- Shearing or cropping (By reducing the length of projecting fibre)
- Brushing (For tighter woven fabric, loose fibre brought to surface)
- Singeing (Sheared and cropped fibre with brushing by partly fuses the fibre)
- Thermosetting (Restrict fibre migration)
- Hot impregnation treatment for synthetic
- Steaming (above 100°C reduce pilling tendancy)
- Milling (Reduced by tighten the fabric structure)
- Thermal shock (Alter fibre properties to restrict pilling)
- High pressure heat treatment with liquid impregnation.
- Ultrasonic treatment (Reduced by modifying fibre surface)

→ Chemical processes
- Treatment based on Silicic acid (Reduce surface friction to minimize entanglement)
- Latex treatment (Improve fibre cohesion and durability)
- Easy care finishes (Enhance fabric stability and wear resistance)
- Shrink-resist treatment (Prevent shrinkage by chlorination or polymer coating)
- Treatment based on thermoplastic vinyl resin (Enhanced fibre bonding to resist abration)

(10)

* Fabric rating after pilling

Rating	Description	Details
5	No pilling	The fabric remain smooth and show no visible pilling.
4	Slight pilling	Small amount of visible pill, not significantly noticable.
3	Moderate pilling	Can having 2 result, 1) Moderate fuzzing or 2) Isolated fully formed pills
4	Significant pilling	Distinct fuzzing
5	Severe pilling	Heavyly covered with pills, uncomfortable and unattractive, unsuitable for use.

Sample preparation → ICI box pilling test procedure → Report

* Sample preparation :-

(1) Like previous one. (Martindale abrasion tester)

(2) Dimension of the specimen - 5"×5" (125mm×125mm)

(3) Like previous one.

Testing Procedure :-

(125 mm x 125 m)

i. Four specimens each of 5" x 5" dimension are cut from fabric. A seam allowance of 12mm is marked on the back of each square.

ii. 2 of the sample seam is marked parralel to the warp direction and other 2 on the weft direction.

iii. Now they are folded face to face and seam is sewn on the marked line.

iv. Each specimen is turned inside out and 6mm cut off each end of it thus removing any sewing distortion.

v. Both ends of the sample were fixed on polyurethane tube

v. Clean the Ici pilling tester and put 4 sample into pilling box, close the lid, start the instrument and rub the sample with the ~~cork~~ cork material.

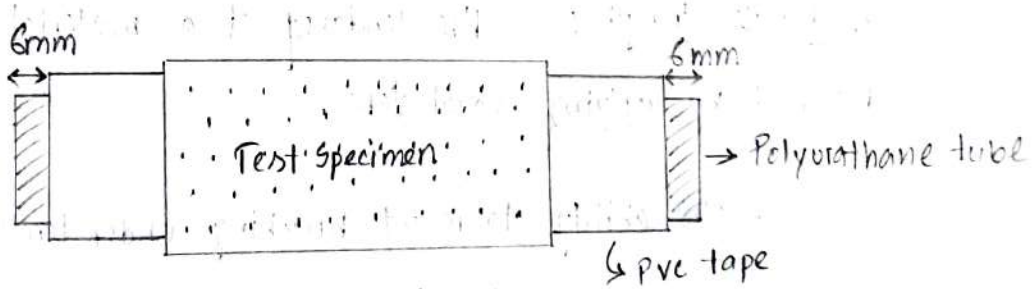


fig: The preparation of pilling sample

vi. There is no specific no of revolution, but for coarse wooven fabric turned over 7200 rpm and for fine wooven it is 14400 rpm.

vii. After the test is completed, the test sample is taken out and compared with orginal one for evaluation.

* Report :- i. This test is conducting according to ISO-12945-1 by using ICI pilling box.

ii. Types of material tested (wooven)

iii. There is 4 number of reading in both warp and weft way direction.

iv. No test is rejected.

—:o:—

ii. Fabric Strength

(a) Tensile strength (N/m) or (MPa)

* Defn + short note

Tensile strength :- The tendancy of a material not to be deformed by applying axial force.

- The ability to resist breaking under tensile stress is one of the most important and widely meanured properties of materials used for industrial application.

- If tensile strength is more than more force is required to deform the material.

Tensile strength test + Sample preparation + Report

Sample preparation :-



i. Like previous one.

ii. Dimension of the speciman :- 300mm x 60mm

iii. Like previous one.

Procedure :-

i. Take 4 sample of the dimension 300mm x 60mm.

ii. After checking and calibrating the tensile strength testing m/c set the test parameter -

① Gauge ~~len~~ Length = 200mm (For fabric with up to 75% stretch)
= 100 mm (" more than 75% ")

② Rate of extension = For gauge length 200mm, 20 mm/min
100mm, 100 mm/min

iii. Mounting the specimen with pretension with the help of bottom and top jaw.

iv. Set the pretension with respect to fabric weight. If

Fabric weighing up to 200g/m² use, 2N force
(200-500) g/m² 5N "
more that 500g/m² 10N "

v. Centre the test specimen in the clamping jaw

vi. Engage tensile testing machine to software (Testwine) and select ISO-13934/1 method for recordinging maximum force and elongation.

vii. Stretch the speciman by moving bottom jaw until it break. Repeat it for all the 4 sample and record the data (Breaking strength)

* Test Report

i. This test is conducted as per ISO-13934/1 by using tensile strength tester.

ii. Types of material tested (woven)

iii. There is 4 number of reading in both warp and weft way.

iv. 2 test sample is rejected due to slippage more than 2mm.

(6) Tear strength (N)

* defn + short note

Tear strength :- Property of a textile material to withstand the effect of tearing or resistance to propagation of tear after it has been initiated.

- This property is important in industrial fabrics that are exposed to rough handling to evaluate britteness and serviceability.

- If tear strength is more of a material it's durability and serviceability is also more.

Sample preparation → Elmendorf tearing tester procedure → Report

* Sample preparation :-

i. Like previous one. Here using a specified cutting die for sample.

ii. Dimension of the speciman :-

100mm x 75mm

iii. Like previous one.

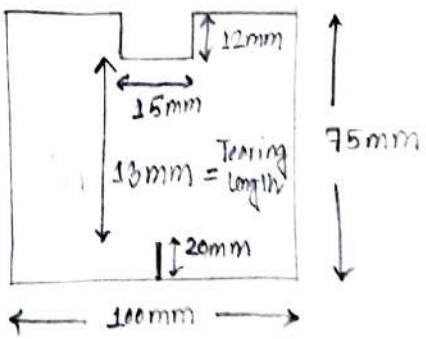


fig: Cutting die

* Procedure :-

i. A sample of dimension 100mm x 75mm is cutting by ~~with~~ cutter using cutting die is taken.

ii. Than sample is placed between the fixed and movable jaw of the tester. after rising and secure the pendulum in the starting position.

iii. An initial slit of 20 mm along the sample is cutted ~~by~~ manually by cutter or cutter mounted on fixed jaw.

iv. Ensure the specimen's edges are parallel to the grip and tighten it.

v. Than release the pendulum to tear the speciman and catch it on return swing without touching pointer.

vi. Record the needle position on the scale where pointer is pointed at.

vii. Repeat it for all specimen and average out the strength.

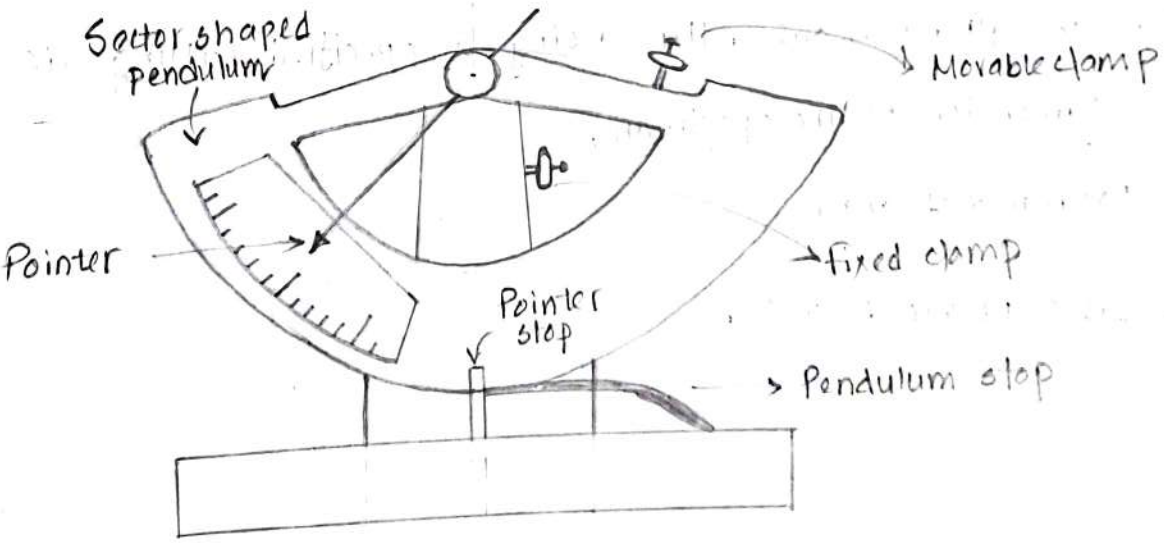


Fig: Elmendrop tear tester

* Report :-

i. This test is conducted according to ISO 13937-1 by Elmendorf tear tester using ballistic pendulum method.

ii. Type of material ~~used~~ tested (wooven)

iii. 4 number of reading in both warp and weft way direction.

iv. 1 sample testing is rejected due to cross-wise tearing.

⑥ Bursting strength (KN/m²)

* Defⁿ + shortnote

Q/A Bursting strength :- Method of measuring strength in which the material is stressed in all directions at the same time.

- Application of a perpendicular force to a fabric until it rupture. The force is normally applied by using either a ball or a hydraulically expended diaphragm.

- Normally brusting strength is used for knitted fabric.

Q/n Sample preparation → Diaphragm brusting test procedure → report

* Sample preparation :—

i. Like previous one. (10 specimen)

ii. Dimension of speciman :—

- 2 sized is used = 30mm and 113 mm

iii. Like previous one.

* Procedure :—

i. Fabric sample is clamped over the rubber diaphragm.

ii. The pressure in the fluid increases at such rate the speciman burts within 20±3 sec. (P_1).

iii. The height (extension) of diaphragm is noted.

iv. Anothe test is carried out without specimen. Pressure to do this is noted. (P_2).

v. ($P_1 - P_2$) is the actual brusting strength of the speciman.

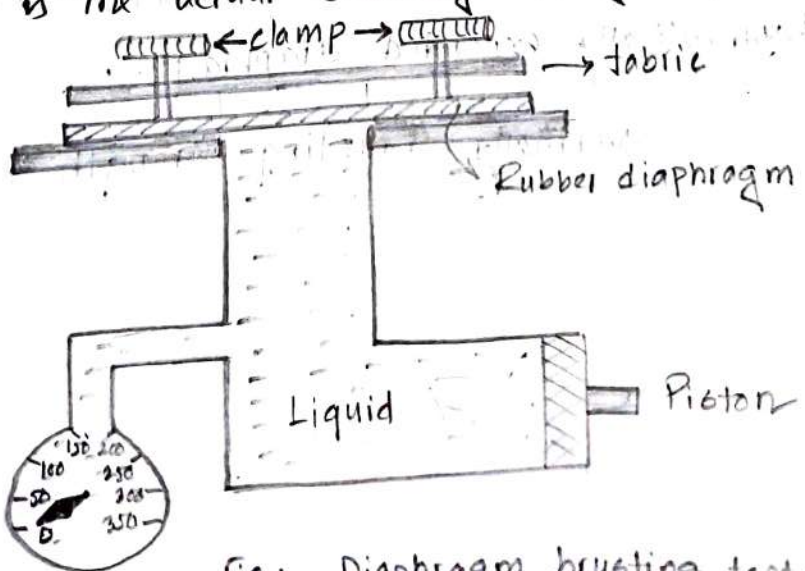


fig :- Diaphragm brusting tester

* Report :-

i. This test is conducted according to ~~D3787~~ ASTM D3787 by diaphragm bursting tester.

ii. Type of material used. (wooven)

iii. No of 10 sample is reading used in both warp and weft way direction.

iv. No sample testing is rejected.

— 0 —

(iii) Fabric Stiffness

* Defⁿ + short note

Ans Fabric Handle :- Refers to the tactile sensations and mechanical properties experienced when fabric is touced, held or manipulated.

- It is a qualitative measure of how the fabric feels to the hand and include characteristic such as softness, stiffness, flexibility, drapability and elasticity.

- Such factor influenced fabric handle like -

1) fibre type
2) Yarn structure
3) Fabric construction
4) finishing processes
5) Environmental condition (Temperature, RH%)

Stiffness :- Tendency of fabric to continue to stand without any support, it is the resistance to bending.

- It is consider the basic features for checking.

i) Fabric suitability
ii) Drapability

- More stiffness means high young modulus also means high bending strength thats result less drapability which makes fabric less comfortness.

- More stiffness required for upholstery fabric and less for apparel. It's property depends on,

i) Fibre type
ii) Yarn twist/inch
iii) Yarn stiffness and diameter
iv) Fabric structure

Bending length :- The measure of draping quality of fabric.

$$\text{Bending length, } c = [L(\cos\tfrac{\theta}{2})^{1/3}]/8\tan\theta, \text{ cm}$$

Here, L = Length of fabric at which strip coincides
$\theta = 41.5°$, standard angle of deflection

- Stiffer fabric having high bending length and a lack of drape and flexibility.

Q. Flexural rigidity :- The measure of stiffness associated with handle is flexural rigidity.

Flexural rigidity, $G = W \times C^3$ gmcm

Here, W = Cloth wt in gm/cm²

C = bending length in cm

Overall flexural rigidity, $G = (G_1 + G_2)^{1/2}$ gmcm

Here, G_1 = flexural rigidity in warp way

G_2 = " " " weft "

- factors like fibre type, yarn structure, fabric construction fabric finish affect flexural rigidity.

Q. Drapability :- Ability of a fabric hang or fold gractully in soft flowing curves.

- It is a key property to determine how fabric comfort to shape of an object or body making it essential for designing garments and home furnishing.

- Factors influencing like flexural rigidity fabric wt, fibre type, weave and knit structure, finish affect drapability.

(2)

* Factors affecting fabric handle, stiffness and drapability

- Dep^n of handle, stiffness and drapability. Than the affecting factors like -

- Fibre type - Varring with diff~~ent~~ types of natural and synthetic fibre.
- Fabric construction - Loose structure having good handle but bad stiffness and drapability.
- Fabric finishes - Fabric finishes like treatment with softeners improve handle, stiffening finishes ~~improve~~ alter stiffness and drapability.
- Yarn structure - Compact or loose structure affect handle, stiffness and drapability in a significant way.
- Others - Less stiffness better handle, drapability and lower flexural rigidity better drapability.

Q/A Sample preparation → Shirley stiffness test procedure → Report

* Sample preparation :-

(1) Like previous one. With help of the template.

(2) Dimension of the specimen :- 200 mm × 25 mm

(3) Like previous one.

Procedure :- (BS 3356)

i. A sample of dimension 200mm x 25mm is cut with cutter with the help of templete.

ii. Bending angle indicator is set at 41.5° as marked on scale.

iii. Movable slide is removed and sample is plaed on horizontal platform, keeping the length at the same parallel to the edge of the platform. The face side of the sample should be up.

iv. Than plaud the movable slide on sample by maintaining sample position.

v. Move the sample manually with hand approximately 120 mm/min so that the edge doesn't touch the knife edge.

vi. Note the overhang length and messure bending length by the formula below -

$$\text{Bending length (cm)} = \frac{\text{Overhang length (cm)}}{2}$$

$$\therefore C = \frac{O}{2}$$

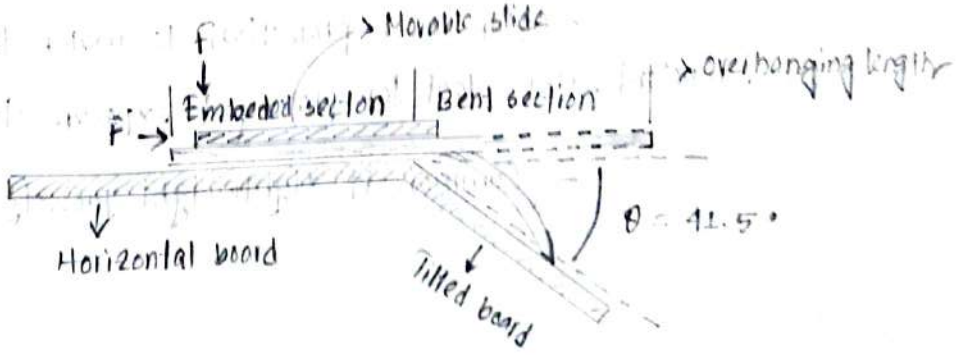


fig: Shirley stiffness tester (cantilever test method)

(iv) Others

* Defn + short note

1/2 Waterproof :- Offer highest level of water protection. Designed to keep water completely out, even during heavy rain.

- It means completely impervious to water, preventing water from ~~prev~~ penetrating of material under any condition.

- Commonly used in outdoor gear, rain jackets and technical equipment. Achieved through coating like Gore-tex.

1/2 Water resistance :- Offer partial resistance to water, repelling it at some extent but not completely blocking it from penetration.

- Suitable for light rain or short exposure to moisture. Achieve though durable water repellent (DWR) coating.

1/2 Showerproof :- Design to resist light rain or short shower but not intended for heavy rain or sustain wet condition.

- Offer lower protection to water than waterproof or water repelant. Not ideal for prolonged exposure to heavy rain.

- Common in lightweight jackets and apparel.

* Relation of stiffness with crease, handle & drape

i. High stiffness leads to better crease resistance, meaning the fabric is less likely to form crease or wrinkle.

ii. A stiffer fabric will generally feel harsher and less soft to the touch compared to less stiff fabric, which have a softer handle.

iii. A stiffer fabric will drape less easily due to it's resistance to bending, while a less stiff fabric ~~having~~ will drape well.

+ - Factors affecting stiffness (P-22)

* Relation between fibre, yarn and fabric strength :-

- Fibre is the base unit of textiles, strongest in raw form.
  - Longer and finner fibre having more strength.
- Yarn is made by twisting fibre, improve tensile strength.
  - Optimal twist, blending and finer ~~fibre~~ yarn having more strength
- Fabric is formed by weaving, knitting or boding yarn.
  - Tighter weave, stronger yarn made fabric stronger.

~~Due to processing state~~

We know that, strength is decreased due to -

i) Processing stage

ii) Distribution

As fabric is made from yarn and yarn is made from fibre, fabric go through more processing stage and disturbance than yarn which results fabric more weaker among them.

As a result, according to strength hierarchy,

Fibre > Yarn > Fabric

—o—